Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/01/2005
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Project Title:	Chemically-Passive Suppression of Laminar Non-premixed (Diffusion) Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-2140	**Congressional District:**	15
Comments:	Faeth, Gerard M.--DECEASED Jan 2005 (per N. Hall/GRC)		
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	02-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Faeth, Gerard M.--DECEASED Jan 2005 (per N. Hall/GRC)		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA94G		
Performance Goal No.:			
Performance Goal Text:			

This research investigates the effects of chemically-passive flame suppressants on the properties of steady laminar opposed-jet diffusion flames at conditions where effects of earth gravity are either small (weakly buoyant flames) or are entirely absent (flames in microgravity environments). Various simple fuels will be considered for computational tractability, as follows: hydrogen due to its broad flammability limits and its very strong resistance to suppression; wet carbon monoxide because its properties are characteristics of conditions where effects of preferential-diffusion are small; and light hydrocarbons, e.g., methane, ethylene and propane, that provide readily controllable simulants of many practical combustible materials (diluting the fuel with an inert gas when necessary in order to avoid the complications of soot formation). Various oxygen/nitrogen mixtures and pressures of 0.1-1.0 atm will be considered to simulate spacecraft environments. Finally, various concentrations of chemically-passive flame suppressants will be considered, to simulate the dilution of flame atmospheres by the action of a fire-extinguishing system, as follows: argon, nitrogen, and carbon dioxide as substances that have been used as chemically-passive flame suppressants for confined environments. The

Task Description:	measurements will be reduced to find effects of test conditions on the structure of the test flames, in order to evaluate numerical computations of flame structure, and to develop simple concepts of flame suppression effectiveness such as minimum oxygen indices for extinction, critical strain rates for extinction and Huggett critical heat capacity of the atmosphere for extinction, that find ready application to the design of practical fire-extinguishing systems. The objectives of this research are: (1) to measure the effect of typical chemically-passive flame suppressants on laminar diffusion flames at conditions where the effects of buoyant motion due to earth gravity are small, i.e., at conditions relevant to spacecraft fire safety, and (2) to use the measurements to evaluate both detailed numerical simulations of the flames as well as simplified concepts of flame suppressant effectiveness such as minimum oxygen indices, critical specific heat capacities of the atmosphere, critical strain rates for extinction, etc., in order to provide rational and readily used design criteria for developing practical fire extinguishing systems for spacecraft. The significance of this research stems from the justification for limiting the study to consideration of effects of chemically-passive suppressants on the properties of laminar diffusion flames. First of all, only chemically-passive suppressants are considered because chemically-active suppressants, such as Halons, generally are not allowable for spacecraft applications due to problems of astronaut health concerns and potential poisoning of spacecraft life support systems. Next, although most practical flames are turbulent, turbulent flames are difficult to study because experimental conditions are substantially complicated by the need to describe turbulence properties. In addition, turbulent flames cause turbulence-induced stretching and quenching, making them less conservative indicators of flame extinction than corresponding laminar flames. Another advantage of laminar flames is that they generally are tractable for detailed numerical simulations, unlike turbulent flames, enhancing capabilities to use computations to supplement directly measured properties. Laminar flames also are relevant to practical turbulent flames based on widely-accepted laminar flamelet concepts of turbulent flames. Finally, due to the complexities of turbulent flames, it seems unlikely that understanding of turbulent flame suppression will precede understanding of laminar flame suppression. Thus, for these reasons, it seems prudent to consider the suppression of laminar flames at the outset.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits both the advancement of fundamental combustion science and the terrestrial fire suppression engineering community and industry.
Task Progress:	Initial numerical simulations of the effects of suppressants on laminar diffusion flames were carried out using the SANDIA code OPDIFF. Reactants of interest include hydrogen, wet carbon monoxide as fuels, with helium, argon, nitrogen, and carbon dioxide as suppressants. Oxidizing atmospheres include air and EVA-preparation atmospheres. These computations considered wet carbon monoxide flames, involving fuel mixtures of carbon monoxide and 3%, 5%, 10%, and 20% (by volume) hydrogen burning in oxidant mixtures of air and 0%, 10%, 20%, and 30% (by volume) suppressant. Initial computations were undertaken to base line present work with the earlier computational and experimental work of Hassan et al. (1997) for the same wet carbon monoxide fuel mixtures but burning in pure air as premixed flames. In this case, the numerical algorithm was the SANDIA code PREMIX involving the wet carbon monoxide/air chemical kinetics of Kim et al. (1994) and Wang and Rogg (1993). The results of these baseline calculations are satisfactory thus far. The test apparatus was assembled and some initial check-out tests were completed considering wet carbon monoxide as the fuel burning in air. These tests were quite successful with flame extinction conditions measured as a function of strain with little difficulty. It was decided to use premixed hydrogen/carbon monoxide fuel mixtures and premixed air/suppressant oxidant mixtures in order to minimize reactant mixture blending errors by flow metering. This is important for the present test series because fuel and oxidant mixture flow rates to the opposed jet diffusion flame apparatus must be increased, while keeping the mixture compositions and the flame position unchanged, until quenching conditions are reached. In particular, initial attempts to do this while controlling four reactant streams (carbon monoxide, hydrogen, air, and suppressant) proved to be very difficult. Finally, all the flow meters were calibrated for the reactant mixtures of interest during the current series of test using wet-gas meters.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Awards	Faeth GM. "National Associate, National Academies, January 2004." Jan-2004
Awards	Faeth GM. "Space Processing Award, American Institute of Aeronautics and Astronautics, January 2004." Jan-2004
Awards	Faeth GM. "Alfred C. Egerton Gold Medal, The Combustion Institute, July 2004." Jul-2004
Awards	Faeth GM. "Arthur B. Modine Distinguished University Professor of Aerospace Engineering, The University of Michigan, September 2004." Sep-2004